

IMFASHANYIGISHO KU BUTAKA, IBIHINGWA NO KURWANYA UDUKOKO



GUPIMA INTUNGABINGWA ZIRI MU UBUTAKA



Intangiriro

Mbere yo gutera, ni ingenzi cyane kumenya niba ubutaka bwawe bufite intungagihingwa zihagije. Gupima ubutaka bikwereka ibikenewe kugira ngo wirinde guta amafaranga meshi ku fumbire ubutaka budakeneye

Igihe cyiza cyo gupima ubutaka

- Birakwiye gufata ibipimo mugihe cy'izuba.
- Bikorwa mbere y'igihe cy'ihinga kugira ngo hamenyekane ingano y' amafumbire azakenerwa mu gihe cy' ihinga. .
- Ni byiza gupima ubutaka nibura buri myaka 4.
- Kumenya amafumbire ubutaka bukeneye
- Kureba niba ubutaka ari ubutaka bw'umunyu (alkaline) cyangwa bufiteubusharire bukabije (acidic)
- Kugenzura uburumbuke bw'ubutaka mbere yo gutera
- Guhitamo ifumbire ikwiye
- Gukoresha ubutaka neza, wirinda gukoresha amafaranga menshi kandi utongera umusaruro

Intambwe ya 1: Menya Impamvu Yo Gupima Ubutaka

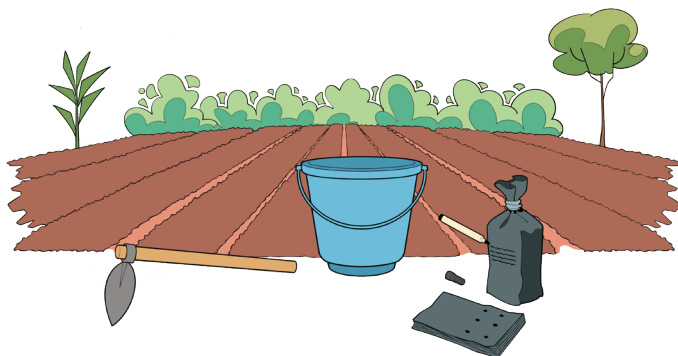
- Guhitamo igihingwa gikwiriye



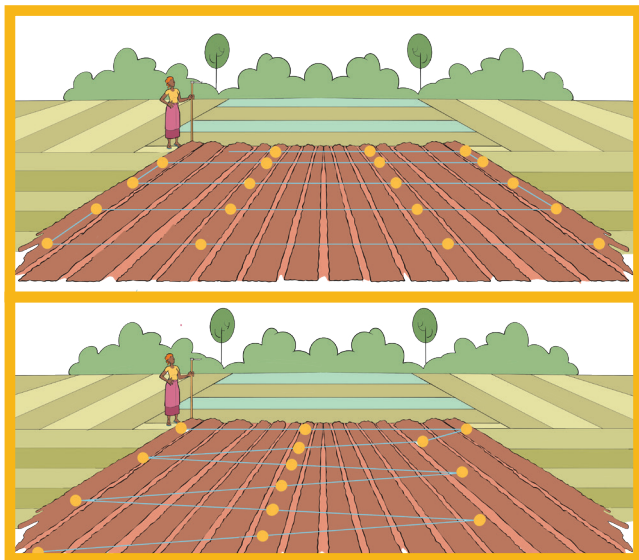
- Ubutaka bwanjye bukeneye iki?
- Nabwongerera umusaruro nte?
- Nkeneye gushyiraho ifumbire yihe? ingana ite?

Intambwe ya 2: Kugabanya umurima mo ibice bito, no gufata ibipimo fatizo by'ubutaka

**Ibikoresho: Indobo, Isuka ntoya , Amasashe
Yabugenewe, Ikaramu Idasibama.**



- Gabanya umurima wawe mo ibice bingana uko bikwiye kugirango ufate ubipimo hose.
- Fata ibipimo muri buri gice wagabanyije, ibyo bizatuma ubasha gufata ibipimo mu murima wose.



- Kuraho ibyatsi, cyangwa ibindi bintu biri hejuru y'ubutaka ugiye gufataho igipimo.
- Cukura umwobo ureshya na 30 cm



- Fata ubutaka ucukuye ubushyire muindobo
- Vanga vanga ubutaka neza
- Fata ibiganza bibiri by'ubutaka bwavanzwe, ubushyire mu ishashi cyangwa ikindi kintu cyagenewe gutwara ibyo bipimo.
- Jya ahandi hantu hagenwe ukomeze ufata ibipimo mu murima wose
- Nurangiza gushyira ubutaka mu ishashi, andika ku ishashi amazina cyangwa imibare igaragaza aho hantu buvuye:

- Amazina yawe
- Nimero cyangwa ID y'igice cy'umurima (Plot 1)
- Nimero y'ibipimo byafashwe.
- Itariki yafatiweho ibyo bipimo.
- Andika numero yihariye y'igipimo cyafashwe ukirikije imyobo wacukuye cyangwa ibipimo ndagahantu byaho hantu byafashwe niba bishoboka.



Intambwe ya 3: Kujoyana ubutaka muri laboratwari

Nyuma yo gukusanya ubutaka, ntutinde kubigeza mu kigo cy'ibizamini ngo bisuzumwe. Hasi hari amwe mu malaboratwari yemewe mu Rwanda. Buri gihe jya ugisha inama cyangwa ubufasha agronome w'umurenge mu bijyanye no gupimisha ubutaka.

1. Laboratwari ya Rubona

📍 Aho iherereye: Huye

☎ Telephone: +250 783 502 100

📦 Igiciro cyo gupima ubutaka gitangirira ku: 21,500 Frw kuri buri cyitegererezo

2. Laboratwari ya NAEB

📍 Aho iherereye: Kigali

☎ Telephone: +250 783 502 100

📦 Igiciro cyo gupima ubutaka gitangirira ku: 36,600 Frw kuri buri gipimo



NB: Jya ubanza kugisha inama agronome w'umurenge cyangwa w'akarere kugira ngo agufashe mu kugeza ubutaka aho bupimishirizwa.

Intambwe ya 4: Gusobanukirwa Ibisubizo by'Ibyavuye mu Bupimire

Agoronome azagufasha gusobanukirwa ibisubizo no gufata umwanzuro kubipimo byapimwe. Ibisubizo bishobora kuza bimeze nkibi bikurikira.

Número	Intungagingwa zapimwe	Ibisobanuro
1	pH (ubusharire bw'ubutaka)	<6 = ubutaka bufite aside nyishi; $6-6.5$ = bwiza; >7 = ubutaka bufite ubusharire bucyeye
2	Ikigero kibinyabuzima biri mubutaka (Organic matter)	$5-8\%$ cyangwa hejuru = ubutaka bwiza
3	Nitrojeni (N)	$50-125$ = Ni byiza
4	Fosifori (Phosphorus)	$30-75$ = Ni byiza ku bihingwa
5	CEC (Ubushobozi bwo gufata intungagihingwa)	>12 = bufata intungagihingwa neza; <8 = (Ubutaka bw'umucanga), bufite ubushobozi bucyeye bwo gufata intungagihingwa.



KUBUNGABUNGA IFUMBIRE Y'UBUTAKA



Intangiriro

Kubungabunga neza ifumbire y'ubutaka ni ingenzi kugira ngo imyaka ibe myiza kandi itange umusaruro mwinshi. Iyi mfashanyigisho igaragaza uburyo bworoshye bwo kubungabunga ifumbire mubutaka bigatuma umwero wiyongera.

Ubuhinzi bubungabunga ubutaka ni bumwe mu buryo bw'imihingire irambye, bufasha umuhinzi kongera umusaruro akoresheje igishoro gicye. Mu byiza by'ubu buhinzi harimo kongera ubuhehere bw'ubutaka, kwihanganira ihindagurika ry'ikirere, no kongera uburumbuke bw'ubutaka.

Impamvu yo Guhitamo Ubuhinzi bushingiye ku kubungabunga ifumbire y' ubutaka

- Bwongera umusaruro, binyuze mu kongera ifumbire karemano y'ubutaka.
- Bugabanya igishoro gikoreshwa mu gutegura ubutaka.
- Burinda ubutaka isuri no gutakaza intungagihingwa.
- Bubika amazi mu butaka bukanagabanya amazi akoreshewa mu gihe bavomerera.
- Bwongera udusimba tw'ingirakamaro (inshuti z'abahinzi) mu butaka.

Inkingi 3 z'Ubuhinzi Bubungabunga Ubutaka

1. Kudahungabanya ubutaka (Minimum Tillage)

- Hinga gusa aho imbuto zizaterwa (utwobo cyangwa uturongo). Bituma utunyabuzima twakamaro tuba mu butaka tudapfa, bugumana ubuhehere, kandi ntibwangirike.



2. Gusasira ubutaka (Permanent Soil Cover)

- Sasira ubutaka ukoresheje ibisigazwa byavuye mu isarura rishize (nk'ibikenyeri by'ibigori cyangwa ibisigazwa by'imyumbati), cyangwa imyaka isasira ubutaka (nk'uruhengeri).
- Bituma ubutaka bugira ubushobozi buhagije bwo kubika amazi, ntibwangirike, kandi ni nuburyo bwiza bwo kwirinda isuri ndetse n'imyuzure.



3. Guhingana no Guhinduranya imyaka (Crop Diversification and Crop rotation)

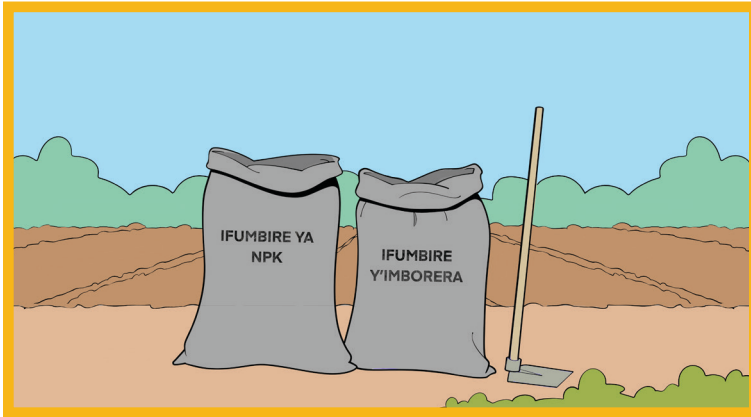
- Hinga imyaka itandukanye niyo wahinze mu gihebwe cy'ihinga gishize cyangwa uhinge hamwe (nko guhinga ibigori n'ibishyimbo by'imishingiro).
- Bituma indwara n'ibyonnyi bigabanuka kandi ifumbire y'ubutaka ikiyongera.



Intambwe mukongerera ubutaka ifumbire n'uboshobozi

Igice cya 1 – Mu gihe cyo Gutera – Gushyiraho Ifumbire ya Mbere (Basal Application)

- Buri gihe kora ihinduranya ry'imyaka, nko guhinga ibishyimbo nyuma y'ibigori. Kugirango bigufashe kubungabunga ifumbire mu butaka bwawe
- Tera ibihingwa bitandukanye urugero: ibigori n' ibishyimbo kugira ngo azote yiyongere mu butaka bwawe kandi ibishyimbo bigufashe no gutwikira ubutaka kugira ngo ubuhehere butindemo.
- Kora imyobo uri buze gushyiramo imbuto maze ubanzemo ifumbire (imborera ndetse ni ifumbire mvaruganda).
- Banzamo ifumbire y' imborera, ikiganza cyuzuye.
- Ugomba kurekera ibyatsi byo gusasira mu murima kugira ngo ubutaka bwawe bugumane ubuhehere kandi iyo bimaze kuma biba ifumbire nabyo.
- Koresha ifumbire mvaruganda nka NPK cyangwa DAP.
 - Ku mwobo umwe, koresha udufuniko 2 tw'icupa rya fanta twa NPK cyangwa DAP
 - Iyi fumbire mvaruganda uyishira mu mwobo umwe ni mborera.
 - Utwikirize agataka gacye, mbere yo guteramo imbuto kugira ngo twirinde ko ifumbire yatwika imbuto.



Igice cya 2 – Gutera Ifumbire ya Nyuma (Top Dressing): Bifasha Igihingwa Gukura nyuma y'Ibyumweru 6 nyuma yo gutera.

Igihe cyo kuyishyiraho

- Bikorwa mu gihe cyo kubagara, nyuma y'ibyumweru 4–6 uhereye igihe watereye.

Impamvu yo kuyikoresha

- Bituma igihingwa gikura neza.
- Gufasha ibimera kuzana uruyange ndetse no kuzana imbuto
- Gukomeza ubuzima bw'igihingwa mbere yo kwera.

Ifumbire isabwa

- Ire (Urea)– ikungahaye kuri azote (nitrogen), ituma igihingwa gikura neza.
- NPK – ifasha imizi gufata ndetse no gukura neza.
- Ifumbire ikungahaye kuri fosifore (DAP cyangwa TSP) – yongera imizi yigihingwa ndetse n'uruyange.

Uko ikoreshwa

- Shyiramo udufuniko 2 twifumbire mvaruganda, ntubikore mugihe hariho izuba ryishi kugira ngo udatwika imyaka.
- Ifumbire iterwa muri centimetero 5–10 uvuye ku gihinga kugirango wirinde yuko wacyangiza.
- Renzaho agataka gake kugirango idatwarwa n’ isuri cyangwa umuyaga.
- Ibi bikorwa bari no kubagara kugira ngo twirinde ko ibyatsi bidakanewe aribyo bifata iyo fumbire.



Igice cya 3 – Igihe intungagihingwa zibaye nke

Hari igihe urubuto rugira intungagihingwa nke, rukagaragaza ibimenyetso bitandukanye. icyo gihe, ifumbire ishyingirwa kuri urwo rubuto gusa, aho kuyishyira mu murima wose.

Mu gihe ibihingwa bitandukanye bigaragaza ibimenyetso bitandukanye, iyi mfashanyigisho yo iribanda ku bimenyetso bigaragazwa n'igihingwa cy'ikigori igihe habaye kubura intungagihingwa. Ifashishe imbonerahamwe ikurikira kugira ngo umenye ibyo bimenyetso, n'icyo wakora igihe bigaragaye:

Ibimenyetso ku Kibabi	Intungagihingwa zabaye nkeya	Ifumbire isabwa
Impande z'ibibabi zisa umuhondo 	Potasiyumu (Potassium)	Udufuniko tw'icupa 2 twa NPK kuri buri gihingwa
Hagati mu kibabi hasa umuhondo 	Azote (Nitrogen)	Udufuniko tw'icupa 2 twa Urea ku gihingwa
Impande z'ibibabi zisa ubururu 	Fosifore (Phosphorus)	Udufuniko tw'icupa 2 twa NPK ku gihingwa

Icyitonderwa: Mu gihe ubonye ibimenyetso nk'ibi by'ayamabara ku bibabi, shyiraho ifumbire isabwa ukoresheje inzira zavuzwe mu gice cya 2.

Inama z'inyongera kugirango w'irinde kandi ukoreshe uburyo bukwiye

 Ibikwiye Gukorwa	 Birabujijwe
Wambara uturindantoki igihe ukoresha ifumbire 	Gutera ifumbire mu gihe hari izuba ryinshi cyangwa imvura nyinshi
Woza intoki nyuma yo gukoresha ifumbire 	Gushyira ifumbire ku gihingwa neza kuko ishobora kugitwika ahubwo yizenyurutse neza kubutaka ahegereye igihingwa 
Bika ifumbire ahantu hakonje kandi humye (ahantu habugenewe)	Gukoresha ifumbire nyinshi birenze urugero
Tera ifumbire igihe imvura irihafi cyangwa imaze kugwa	Gutera ifumbire igihe ubutaka bumaze kuma cyane

GUTEGURA UBUTAKA NO GUHINGA IBISHYIMBO



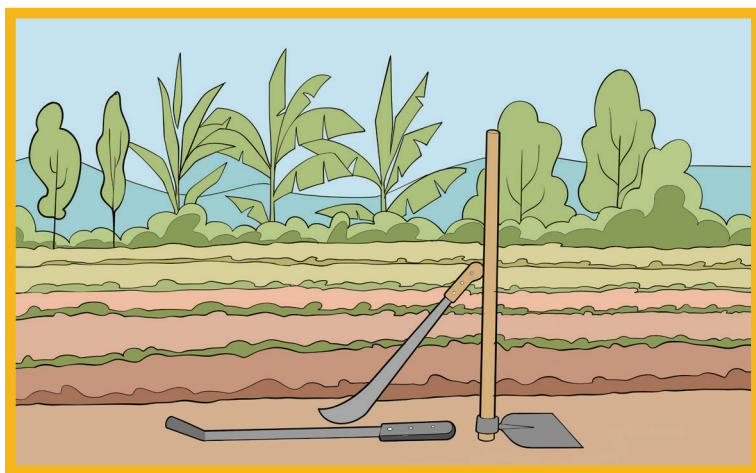
Intangiriro

Gutegura umurima neza no gutera uko bikwiye ni ingenzi mu kongera umusaruro w'ibishyimbo. Iyi mfashanyigisho itanga amabwiriza y'ingenzi yo gukurikiza mu gutegura umurima no gutera, hakoreshejwe amahame y'ubuhinzi burengera ibidukikije kandi burambye.

Gukurikiza aya mabwiriza, bizafasha umuhinzi kongera ireme ry'ubutaka, kubika amazi mu butaka, kugabanya ibyatsi byonona imyaka no kongera umusaruro, byose bikoze hifashishijwe ibikoresho biciriritse abahinzi basanganywe.

Gutegura umurima

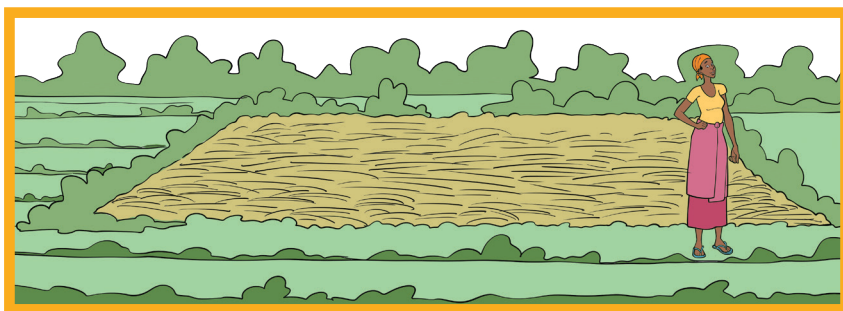
Ibikoresho bikenerwa: umuhoro cyangwa kupakupa, hamwe n'isuka,



- Habura ibyumweru 2 kugeza kuri 4 mbere yo gutera, tema ubwatsi, ibigunda, n'ibisigazwa by' imyaka yahinzwe mu gihembwe cy'ihinga cyashize ukoresheje umuhoro, kupakupa, isuka, cyangwa ikindi gikoresho ufite.

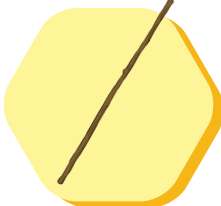
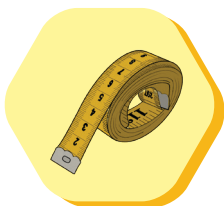
- Ibyatsi wakase binyanyagize mu murima bitwikire ubutaka bwose. Irinde kubikoramo ikirundo, cyangwa se kubitwika. Nibimara kuma, birekere mu murima bibe isaso. Iyi saso, ituma ubutaka buhehera, ikagabanya ibyatsi by'ibyonnyi,
- kandi ikarinda isuri.

Zirikana: Irinde gutwika ubwatsi wakase cyangwa kubukura mu murima. Ubu bwatsi ni isaso yongera uburumbuke bw'ubutaka.



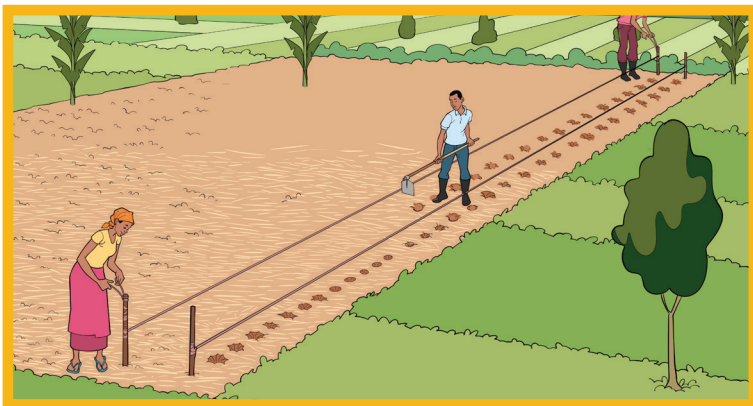
Gutera

Ibikoresho bikenerwa: isuka, ikamba, imetero ipima uburebure, n'imambo



Intambwe ya 1: Shyiraho umurongo wo guteraho

- Shinga ibiti bibiri ku mpera zombi z'aho umurima wawe utangirira, ubizirikeho umugozi. Uwo mugenzi werekana aho umurongo wa mbere w'ibishyimbo uri buterwe.
- Ukoresha imetero, pima inkoni ya cm 40, uyikate (iyi nkoni izajya ifasha gupima intera ijya hagati y' umurongo nuwundi).
- Uhereye ku murongo wa mbere, upime 40 cm ukoresha ya nkoni wakase, ushyireho umurongo wa kabiri. (Buri gihe ukoresha umugozi kugira ngo imirongo ijyane neza kandi intera hagati yayo ibe 40 cm).



Intambwe ya 2: Kora igikoresho cyo gupima intera ijya hagati y'igishyimbo n'ikindi

- Ukoresheje imetero, pima inkoni ya **sentimetero 20**, uyikate aho igipimo kigarukira (iyi nkoni izakoreshwa mu gupima intera y'imyobo yo guteramo imbuto mu mirongo).
- Ku mirongo ibiri wakoze igaragazwa n'imigozi, cukura utwobo duto dufite nka **sentimetero 10 z'ubujyakuzimu na sentimetero 10 z'ubugari**. Hagati y'akobo n'akandi sigamo intera ingana na sentimetero 35 ukoresheje ya nkoni wakase na yo ifite sentimetero 20.



Intambwe ya 3: Shyira ifumbire n'imbuto mu mwobo wakoze.

- Mu mwobo wose, shyiramo ibiganza bibiri by'ifumbire y'imborera (imaze kubora neza).
- Ongeramo udupfundikizo tubiri tw'ifumbire mvaruganda. Udupfundikizo dukoresheya ni utw' icupa ry'amazi ritoya.
- Twikiriza agataka gake hejuru y'ifumbire, muri buri mwobo. Shyira imbuto ebyiri z'ibishyimbo hejuru y'icyo gitaka.
- Renza akandi gataka gacye kuri izo umbuto.



Intambwe ya 4: Subiramo intabwe ya mbere kugeza kuya gatatu, kugeza ubwo umurima wose uwuteye imbuto.

Nyuma yo Gutera

- Sura umurima wawe nyuma y'iminsi 7-10 uteye. Nihaba hari aho imbuto itameze, ihutire kongera kuhatera.



Zirikana:

☒ Uge ukora ibikurikira:	☐ Birabujijwe
● Siga ibisigazwa by'ibihingwa byo mu gihembwe cy'ihinga cyashize ku murima, bibe isaso. ● Hinga gusa ahazajya imbuto, ukoze imyobo nk'uko iyi mfashanyigisho yabigaragaje. ● Tangira gukoresha ubu buryo bw'imihingire ku murima muto, nibigenda neza, wagure ubukoreshe no ku mirima yose ufite.	● Gutwika ibisigazwa by'ibihingwa byahinzwe mu gihembwe cy'ihinga cyashize. ● Guhinga umurima wose.

GUTEGURA UBUTAKA NO GUTERA IBIGORI



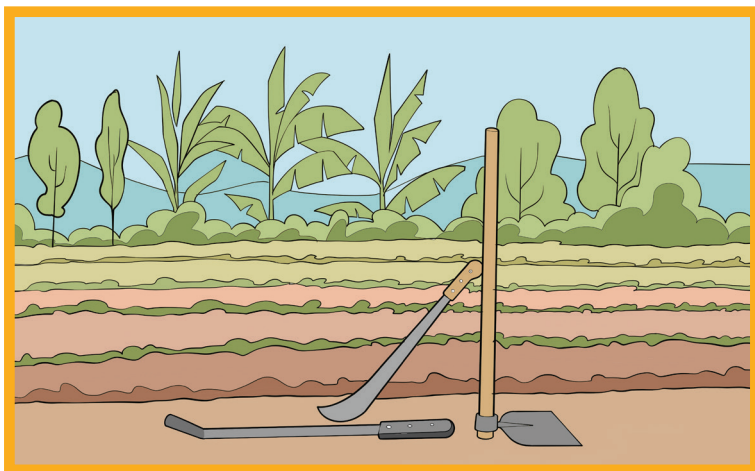
Intangiriro

Gutunganya umurima neza no gutera uko bikwiye ni ingenzi mu kongera umusaruro w'ibigori. Iyi mfashanyigisho itanga amabwiriza y'ingenzi yo gukurikiza mu gutegura umurima no gutera, hakoreshejwe amahame y'ubuhinzi burengera ibidukikije kandi burambye.

Gukurikiza aya mabwiriza, bizafasha umuhinzi kongera ireme ry'ubutaka, kubika amazi mu butaka, kugabanya ibyatsi byonona imyaka no kongera umusaruro, byose bikoze hifashishijwe ibikoresho biciriritse abahinzi basanganywe.

Gutegura umurima

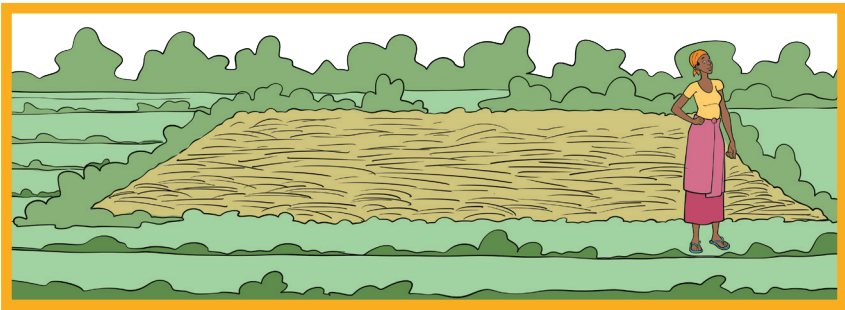
Ibikoresho bikenerwa: umuhoro cyangwa kupakupa, cyanga isuka.



- Habura ibyumweru 2 kugeza kuri 4 mbere yo gutera, harura ibyatsi n'ibisigazwa by' imyaka yahinzwe mu gihembwe cy'ihinga cyashize ukoresheje umuhoro, kupakupa, isuka, cyangwa ikindi gikoresho gikata ufite.

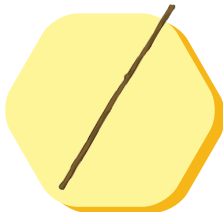
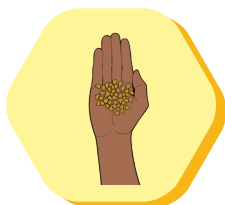
- Ibyatsi wakase binyanyagize mu murima bitwikire ubutaka bwose. Irinde kubikoramo ikirundo, cyangwa se kubitwika.
- Nibimara kuma, birekere mu murima bibe isaso. Iyi saso, ituma ubutaka buhehera, ikagabanya ibyatsi by'ibyonnyi, kandi ikarinda isuri.

Zirikana: Irinde gutwika ubwatsi wakase cyangwa kubukura mu murima. Ubu bwatsi ni isaso yongera uburumbuke bw'ubutaka.



Gutera

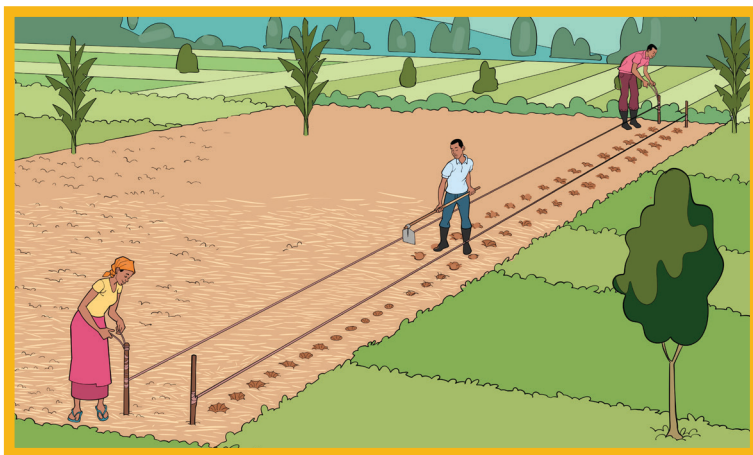
Ibikoresho bikenerwa: imbuto nziza, imigozi ya mushipiri cyangwa indi iboneka, ibiti cyangwa inkoni nk'izikoreshwa mu guhembera, Imetero cyangwa ikindi gikoreshe gihari cyakoreshwa mu gupima uburebure.



Intambwe ya 1: Shyiraho umurongo wo guteraho

- Shinga ibiti bibiri ku mpera zombi z'aho umurima wawe utangirira, ubizirikeho umugozi. Uwo mugenzi werekana aho umurongo wa mbere w'ibigori uri buterwe.
- Ukoresheje imetero, pima inkoni ya cm 75, uyikate (iyi nkoni izajya ifasha gupima intera ijya hagati y' umurongo nuwundi).

- Uhereye ku murongo wa mbere, upime 75 cm ukoresheje ya nkoni wakase, ushyireho umurongo wa kabiri. (Buri gihe ukoreshe umugozi kugira ngo uterere ku mirongo kandi intera hagati yayo ibe 75 cm).



Intambwe ya 2: Kora igikoresho cyo gupima intera ijya hagati y'ikigori n'ikindi

- Ukoresheje imetero, pima inkoni ya **sentimetero 35**, uyikate aho igipimo kigarukira (iyi nkoni izakoreshwa mu gupima intera izajya hagati y'umwobo wo guteramo 1 nuwundi.).
- Ku mirongo ibiri wakoze igaragazwa n'imigozi, cukura utwobo duto dufite nka **sentimetero 10 z'ubujyakuzimu na sentimetero 10 z'ubugari**. Hagati y'akobo n'akandi sigamo intera ingana na sentimetero 35 ukoresheje ya nkoni wakase ifite sentimetero 35.



Intambwe ya 3: Shyira ifumbire n'imbuto mu myobo wakoze.

- Mu mwobo yose, shyiramo ibiganza bibiri by'ifumbire y'imborera (imaze kubora neza).
- Ongeramo udupfundikizo tubiri tw'ifumbire mvaruganda. Udupfundikizo dukoresheya ni utw' icupa ry'amazi ritoya.
- Twikiriza agataka gake hejuru y'ifumbire, muri buri mwobo.
- Shyira imbuto ebyiri z'ibigori hejuru yako gataka.
- Renza akandi gataka gacye kuri izo umbuto.



Intambwe ya 4: Subiramo intabwe ya mbere kugeza kuya gatatu, kugeza ubwo umurima wose uwuteye imbuto.

Nyuma yo Gutera

- Sura umurima wawe nyuma y'iminsi 7-10 uteye. Nihaba hari aho imbuto itameze, ihutire kongera kuhatera.



Zirikana:

☒ Uge ukora ibikurikira:	☐ Birabujijwe
● Siga ibisigazwa by'ibihingwa byo mu gihembwe cy'ihinga cyashize ku murima, bibe isaso. ● Hinga gusa ahazajya imbuto, ukoze imyobo nk'uko iyi mfashanyigishye yabigaragaje. ● Tangira gukoresha ubu buryo bw'imihingire ku murima muto, nubigenda neza, wagure ubukoreshe no ku mirima yose ufite.	● Gutwika ibisigazwa by'ibihingwa byahinzwe mu gihembwe cy'ihinga cyashize. ● Guhinga umurima wose.

KURWANYA IBYONNYI MU BIGORI



Intangiriro

Mu guhangana n' ibyonnyi, dushyira imbere ingamba zo kubikumira, kubyirinda, no kubirwanya. Iyi mfashanyigisho igaruka cyane ku ngamba zo guhangana n'ibyonnyi muri rusange, igatanga urugero kuri nkongwa idasanzwe ikunda kwibasira ibigori igatera igihombo.

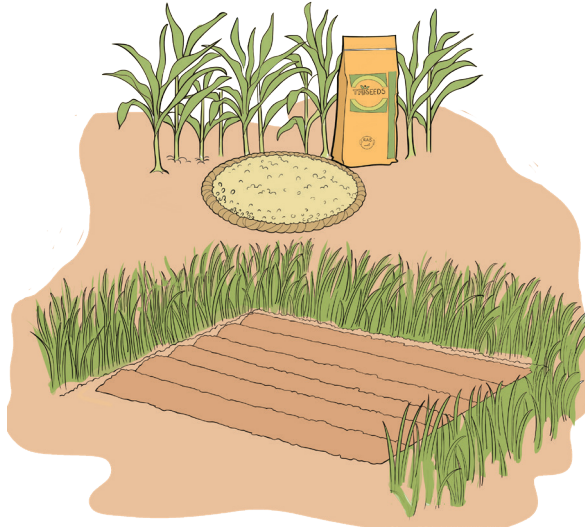
Uburyo bwo kurwanya ibyonnyi

1. Gukumira

Gufata ingamba zo gukumira bigabanya ibyago byo kwibasirwa n'ibyonnyi mu murima.

Inzira zo gukumira:

- Hinduranya ibihingwa: Hinga ibigori usimburanye n'ibishyimbo cyangwa soya.
- Koresha imbuto zemewe n'inzego zibishinzwe, zituruka ku bacuruzi bizewe.
- Sukura umurima: tema ibigunda bikikije umurima muri metero 1 kugeza kuri metero 3 uhaye aho umurima wawe urangirira.
- Hinga ibigori ubivanga na mucuna, kandi uginge uruzitiro rw' urubingo ku nkengero z'umurima.



2. Kurinda

Izi ngamba zigamije gutahura no kugabanya umubare w'ibyonnyi hakiri kare, mbere y'uko bigira ingaruka ku musaruro.

Inzira zo kurinda:

- Sura umurima nibura inshuro ebyiri mu cyumweru nyuma y'uko imbuto zimaze kumera, ushakisha ibimenyetso by'ibyonnyi nk' imyenge y'amababi n'ibisigazwa by'ibyonnyi.
- Niba hari ibyonnyi ubonye, byice.
- Shyira mu murima imitego ya feromone ifasha gufata udusimba e, bityo bikagabanya umubare no kororoka kwabyo.



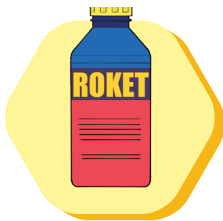
3. Kurwanya

Niba ubona ko 20% cyangwa birenze by'ibihingwa byangiritse bitewe n'ibyonnyi, ni ngombwa gukoresha umuti ubyica. Igice gikurikira kiribanda ku ntabwe zo kurwanya nkongwa idasanzwe, ikunda kwibasira ibigori.

Kurikiza izi ntambwe zikurikira mu gukoresha umuti mu kurwanya nkongwa yibasira ibigori (Fall Army Worm - FAW)

Ibikoresho bikenewe:

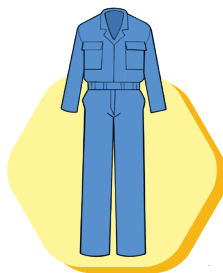
- Umuti wa Rocket wemewe n'inzeho zibishinzwe.
- Ipombo (knapsack sprayer) ifite ubushobozi bwa litiro 15.
- Ibikoresho byo kwirinda: amadarubindi yo kurinda amaso, inkweto zabugenewe (Bote), agapfukamunwa, isarubeti cyangwa indi myenda ifite amaboko maremare n'amaguru ndetse na ga zo kurinda intoki.
- Amazi meza (litiro 15).
- Ibikoresho byo gupima umuti



**Umuti wa
Rocket**



Ipombo



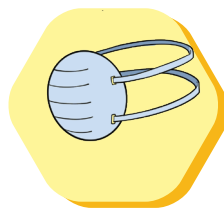
Isarubeti



**amadarubindi yo
kurinda amaso**



**Amazi meza
(litiro 15)**



Agapfukamunwa



**Inkweto
Zabugenewe**

Inama zagufasha kwirinda mugihe urimo gutera umuti

Iyo ukoresha imiti yica ibyonnyi, jya uha agaciro ubuzima bwawe:

- Ambara ibikoresho byo kwirinda: amadarubindi yo kurinda amaso, Inkweto zabugenewe (Bote) agapfukamunwa cyangwa mask, ibirinda intoki, n'igisarubeti cyabugenewe.



Intabwe zo gukoresha umuti wa Rocket wica urwanya nkongwa.

1. Uzuza ipombo yawe amazi meza angana na litiro 15.
2. Ongeramo mililitiro 45 z'umuti wa Rocket (ni nk'udupfundikizo tubiri n'igice tw'icupa ry'uyu muti). Niba utizeye neza ibipimo, egera agaronome ukwegereye cyangwa umucuruzi w'inyongeramusaruro aho waguriye umuti wa rocket, cyangwa umufashamyumvire mu mudugudu kugira ngo wirinde gutera umuti mwishi bityo ukaba wakwangiza ibihingwa.
3. Tera umuti mu murima wose mu buryo bungana ugenda buhoro buhoro, wita cyane ku bice byangijwe na nkongwa. .
4. Ushyire umuti ku mababi, no mu mwuma w'ikigori aho nkongwa ishobora kwihisha.
5. Kora ku buryo umurima wose uwutera umuti.
6. Nyuma y'iminsi itatu, suzuma umurima nanone. Niba ubona ibimenyetso bya nkongwa, tera umuti bwa kabiri.



Nyuma yo Gutera Umuri

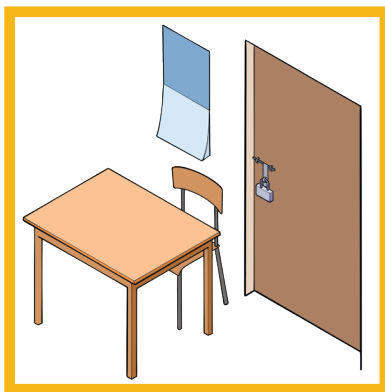
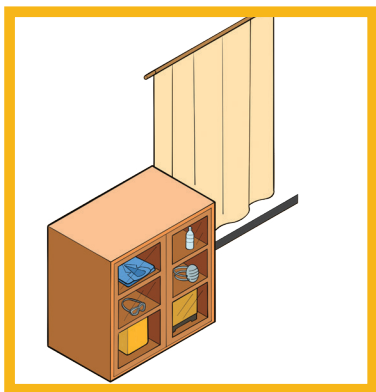
- Sukura ipompo n'amazi meza inshuro eshatu mbere yo kuyibika.



- **Amazi wasukurishije ipombo yamene mu murima wateyemo umuti, ku butaka kugira ngo wirinde kwanduza ibidukikije.**



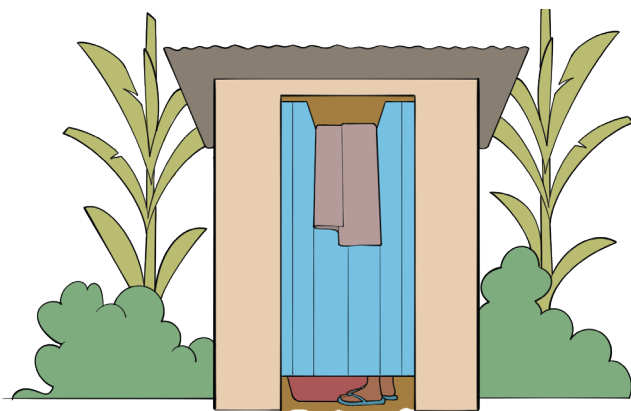
- **Bika imiti mu kabati gafunze cyangwa mu cyumba gifunze, kure y'ibiribwa n'abana.**



- Sukura imyambaro (imyenda&inkweto) wambaye igihe wateraga umuti.



- Iyuhagire umubiri wose



 Uge ukora ibikurikira:	 Birabujijwe
● Mesa imyambaro (imyenda&inkweto) yawe nyuma yo gutera umuti ● Karaba intoki n'umubiri wose nyuma yo gutera umuti ● Igihe utera umuti, genda witonze mu buryo bungana kugeza urangije umurima wose.	● Kwambara imyenda y'amaboko/ amaguru bigufi. ● Gutera umuti utambaye inkweto zabugenewe (Bote) ● Gutera umuti utambaye amadarubindi yabugenewe ● Gutera umuti utambaye agapfukamunwa ● Gusubira mu murima mbere y'amasaha 48 uteye umuti

Zirikana: Gufata ingamba zavuzwe hakiri kare bikumira ibyonnyi, bikongera umusaruro, n'inyungu.

GUPIMA UMUPIRA W' IPOMBO NIBA UKORA NEZA



Gupima neza umupira w'ipombo ni intambwe y'ingenzi mbere yo gutangira igikorwa cyo gutera imiti yica udukoko n'indwara.

Gupima neza bituma imiti isukwa mu rugero rukwiye, ku buryo bungana, bikongera ubushobozi bwayo bwo kurwanya udukoko no kwirinda guta igihe n'amafaranga. Umupira w' ipombo udafunze cyangwa wangiritse ushobora gutuma umuti udaterwaneza, ukaba mucye cyangwa mwinshi. Ibyo bishobora gutera ibibazo mu kurwanya udukoko, kwangiza imyaka, no gutakaza amafaranga. Iyi mfashanyigisho igaragaza intambwe ku yindi uko wagenzura umupira y'ipombo mbere yo kuwukoresha.

Intambwe zo Kugenzura umupira w'ipombo

1. Genzura umupira w'ipombo:

- **Uzuza ipombo amazi meza gusa (nta muti urimo).**



- **Suka amazi ku buso buto mu murima cyangwa hasi kugira ngo urebe uko amazi asohoka ava mu ipombo.**



- **Reba uko amazi asohoka:** Amazi agomba gusohoka angana kandi mu buryo buhoraho
- **Niba amazi asohoka ari udutonyanga duke, cyangwa tugasohoka uko twiboneye cyangwa ntasohoke, bishobora kwerekana ko umupira udafunze neza, cyangwa wangiritse.**

2.Niba umupira wifunze, kora ibi bikurikira:

- **Yihanagure:** Koza umupira ukoresheje uburoso n'amazi kugira ngo ukureho imyanda cyangwa ibisigazwa by'imiti.
- **Gerageza Urebe Impinduka:** Suka amazi uhandi urebe niba noneho umupira usohora amazi nk'dutonyanga duto.
- **Niba nta mpinduka:**

Kuramo umipira w'ipombo uwusuzume urebe ko wangiritse cyangwa ufite imyenge.



- **Niba warangiritse cyane cyangwa ufite imyenge:** Shaka undi mupira usimbura uwo wangiritse. Ushobora kuwubona ku iduka ricuruza ibikoresho by'ubuhinzi (agro-dealer shop).



Icyitonderwa:

- Buri gihe pima umupira w'ipombo mbere yo gutera imiti.
- Koresha amazi meza mu gupima kugira ngo wirinde kuba wagira ingaruka ziturutse n'imiti.
- Bika imipira yasukuwe.
- Ntukigere ukoresha ibikoresho bikarishye mu gusukura umupira kuko bishobora kuwangiza burundu.

